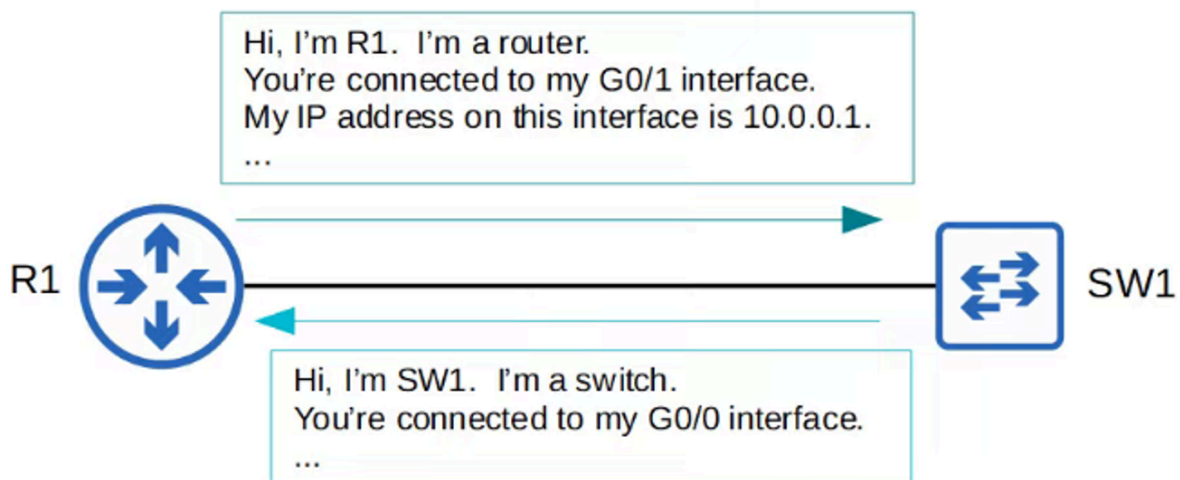


Day 36: CDP & LLDP

INTRO TO LAYER 2 DISCOVERY PROTOCOLS

- LAYER 2 DISCOVERY PROTOCOL, such as CDP and LLDP share information WITH and DISCOVER information about NEIGHBORING (Connected) DEVICES
- The SHARED INFORMATION includes:
 - Hostname
 - IP Address
 - Device Type
 - etcetera.
- **CDP** is a Cisco Proprietary Protocol
- **LLDP** is an Industry Standard Protocol (IEEE 802.1AB)
- Because they SHARE INFORMATION about the DEVICES in the NETWORK, they can be considered a security risk and are often NOT used. It is up to the NETWORK ENGINEER / ADMIN to decide if they want to use them in the NETWORK or not.

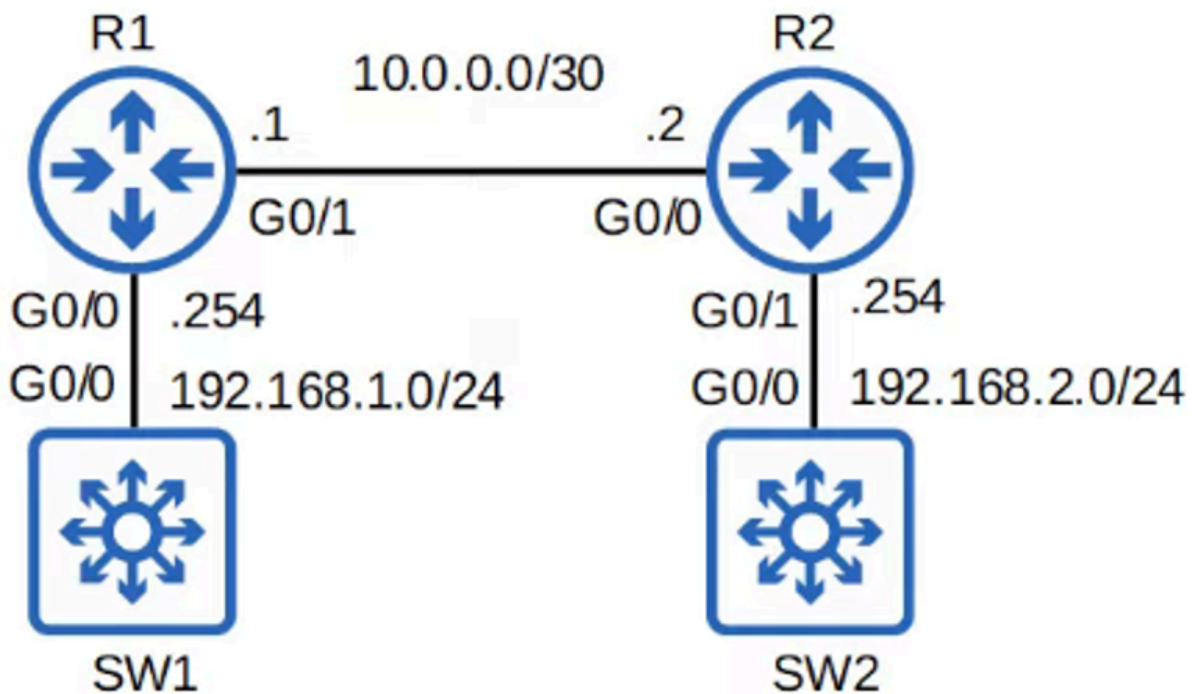


CISCO DISCOVERY PROTOCOL (CDP)

- CDP is a Cisco proprietary protocol
- It is enabled on Cisco devices (routers, switches, firewalls, IP Phones, etc) by DEFAULT

💡 CDP Messages are periodically sent to Multicast MAC ADDRESS
`0100.0CCC.CCCC`

- When a DEVICE receives a CDP message, it PROCESSES and DISCARDS the message. It does NOT forward it to other devices.
- By DEFAULT, CDP Messages are sent once every **60 seconds**
- By DEFAULT, the CDP hold-time is **180 seconds**. If a message isn't received from a neighbor for 180 seconds, the neighbor is REMOVED from the CDP Neighbor Table
- CDPv2 messages are sent by DEFAULT



```
R1#show cdp
Global CDP information:
    Sending CDP packets every 60 seconds
    Sending a holdtime value of 180 seconds
    Sending CDPv2 advertisements is enabled

R1#
R1#show cdp traffic
CDP counters :
    Total packets output: 105, Input: 112
    Hdr syntax: 0, Chksum error: 0, Encaps failed: 0
    No memory: 0, Invalid packet: 0,
    CDP version 1 advertisements output: 0, Input: 0
    CDP version 2 advertisements output: 105, Input: 112

R1#
```

```
R1#show cdp interface
GigabitEthernet0/0 is up, line protocol is up
    Encapsulation ARPA
    Sending CDP packets every 60 seconds
    Holdtime is 180 seconds
GigabitEthernet0/1 is up, line protocol is up
    Encapsulation ARPA
    Sending CDP packets every 60 seconds
    Holdtime is 180 seconds
GigabitEthernet0/2 is administratively down, line protocol is down
    Encapsulation ARPA
    Sending CDP packets every 60 seconds
    Holdtime is 180 seconds
GigabitEthernet0/3 is administratively down, line protocol is down
    Encapsulation ARPA
    Sending CDP packets every 60 seconds
    Holdtime is 180 seconds

    cdp enabled interfaces : 4
    interfaces up          : 2
    interfaces down        : 2
```

CDP NEIGHBOR TABLES

```

R1#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone,
                  D - Remote, C - CVTA, M - Two-port Mac Relay

Device ID         Local Intrfce   Holdtme    Capability   Platform   Port ID
SW1               Gig 0/0        153        R S I        Gig 0/0
R2               Gig 0/1        146        R B          Gig 0/0

Total cdp entries displayed : 2
R1#

```

"Device ID" = What devices were DISCOVERED by CDP

"Local Intrface" = What LOCAL device interface the neighbors are connected to

"Holdtime" = Hold-time countdown in seconds (0 = device removed from table)

"Capabilities" = Refers to Capability Codes table (located above output)

"Platform" = Displays the MODEL of the Neighbor Device

"Port ID" = Neighbor ports that LOCAL device is connected to

MORE DETAILED OUTPUT

```

R1#show cdp neighbors detail
Device ID: SW1
Entry address(es):
Platform: Cisco , Capabilities: Router Switch IGMP
Interface: GigabitEthernet0/0, Port ID (outgoing port): GigabitEthernet0/0
Holdtime : 174 sec

Version :
Cisco IOS Software, vios_l2 Software (vios_l2-ADVENTERPRISEK9-M), Version 15.2(4.0.55)E, TEST ENGINEERING ESTG_WEEKLY BUILD, synced to END_OF_FLO_ISP
Technical Support: http://www.cisco.com/techsupport
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Compiled Tue 28-Jul-15 18:52 by sasyamal

advertisement version: 2
VTP Management Domain: ''
Native VLAN: 1
Duplex: full

-----
Device ID: R2
Entry address(es):
  IP address: 10.0.0.2
Platform: Cisco , Capabilities: Router Source-Route-Bridge
Interface: GigabitEthernet0/1, Port ID (outgoing port): GigabitEthernet0/0
Holdtime : 163 sec

Version :
Cisco IOS Software, IOSv Software (VIOS-ADVENTERPRISEK9-M), Version 15.6(2)T, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
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Compiled Tue 22-Mar-16 16:19 by prod_rel_team

advertisement version: 2
Duplex: full
Management address(es):
  IP address: 10.0.0.2

Total cdp entries displayed : 2

```

"Version" = shows what version of Cisco's IOS is running on the device

SHOW SPECIFIC CDP NEIGHBOR ENTRY

```
R1#show cdp entry R2
```

```
-----  
Device ID: R2  
Entry address(es):  
  IP address: 10.0.0.2  
Platform: Cisco , Capabilities: Router Source-Route-Bridge  
Interface: GigabitEthernet0/1, Port ID (outgoing port): GigabitEthernet0/0  
Holdtime : 178 sec  
  
Version :  
Cisco IOS Software, IOSv Software (VIOS-ADVENTERPRISEK9-M), Version 15.6(2)T, RELEASE SOFTWARE (fc2)  
Technical Support: http://www.cisco.com/techsupport  
Copyright (c) 1986-2016 by Cisco Systems, Inc.  
Compiled Tue 22-Mar-16 16:19 by prod_rel_team  
  
advertisement version: 2  
Duplex: full  
Management address(es):  
  IP address: 10.0.0.2
```

CDP CONFIGURATION COMMANDS



CDP show commands summary

- **R1# show cdp**
→ shows basic information about CDP (timers, version)
- **R1# show cdp traffic**
→ displays how many CDP messages have been sent and received
- **R1# show cdp interface**
→ displays which interfaces CDP is enabled on
- **R1# show cdp neighbors**
→ lists CDP neighbors and some basic information about each neighbor
- **R1# show cdp neighbors detail**
→ lists each CDP neighbor with more detailed information
- **R1# show cdp entry name**
→ displays the same info as above, but for the specified neighbor only

- CDP is GLOBALLY ENABLED, by DEFAULT
 - CDP is also ENABLED on each INTERFACE, by DEFAULT
 - To ENABLE / DISABLE CDP globally: `R1(config)# [no] cdp run`
 - To ENABLE / DISABLE CDP on specific interfaces : `R1(config-if)# [no] cdp enable`
 - Configure the CDP timer: `R1(config)# cdp time *seconds*`
 - Configure the CDP holdtime: `R1(config)# cdp holdtime *seconds*`
 - ENABLE / DISABLE CDPv2: `R1(config)# [no] cdp advertise-v2`
-

LINK LAYER DISCOVERY PROTOCOL (LLDP)

- LLDP is an INDUSTRY STANDARD PROTOCOL (IEEE 802.1AB)
- It is usually DISABLED on Cisco devices, by DEFAULT, so it must be manually ENABLED
- A device can run CDP and LLDP at the same time



LLDP Messages are periodically sent to Multicast MAC ADDRESS
`0180.c200.000E`

- When a DEVICE receives an LLDP message, it PROCESSES and DISCARDS the message. It does NOT forward it to OTHER DEVICES
 - By DEFAULT, LLDP Messages are sent once every **30 seconds**
 - By DEFAULT, LLDP Holdtime is **120 seconds**
 - LLDP has an additional timer called the 'reinitialization delay'
 - If LLDP is ENABLED (Globally or on an INTERFACE), this TIMER will DELAY the actual initialization of LLDP (**2 seconds**, by DEFAULT)
-

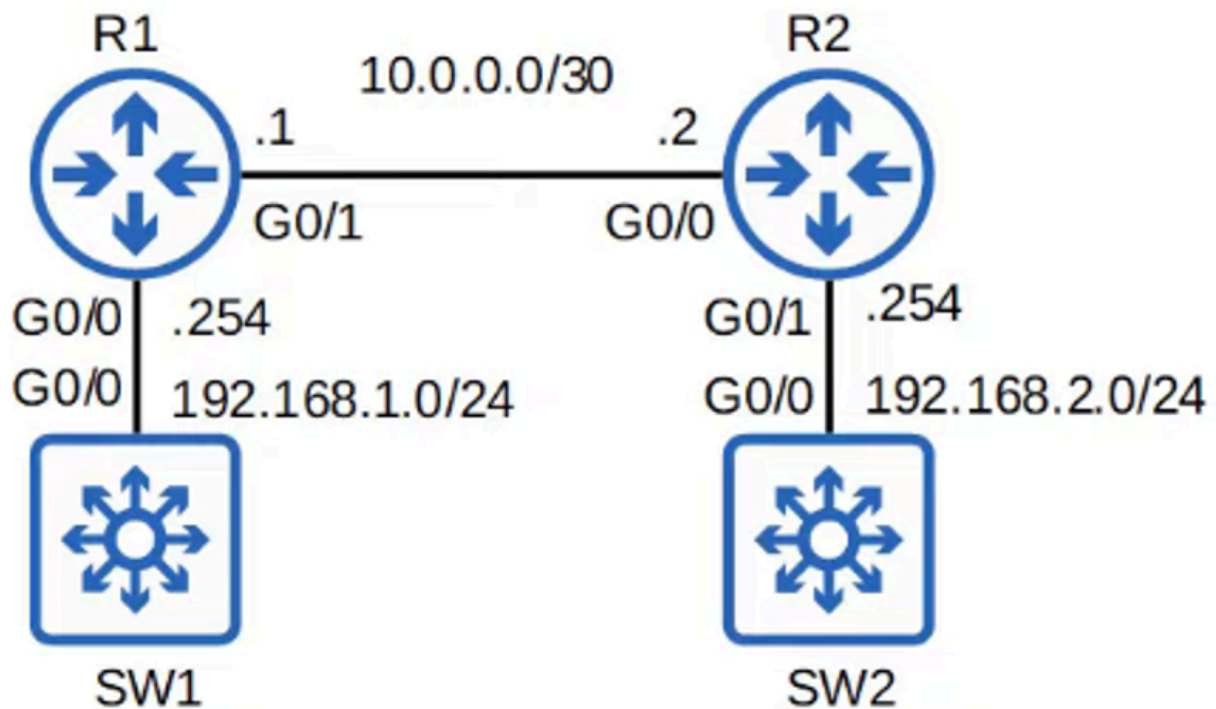
LLDP CONFIGURATION COMMANDS

- LLDP is usually GLOBALLY DISABLED by DEFAULT
- LLDP is also DISABLED on each INTERFACE, by DEFAULT
- To ENABLE LLDP GLOBALLY : `R1(config)# lldp run`
- To ENABLE LLDP on specific INTERFACES (tx): `R1(config-if)# lldp transmit`

- To ENABLE LLDP on specific INTERFACES (rx): `R1(config-if)# lldp receive`

YOU NEED TO ENABLE BOTH TO SEND AND RECEIVE (Unless you want to only enable SEND or RECEIVE LLDP Messages)

- Configure the LLDP timer: `R1(config)# lldp timer *seconds*`
- Configure the LLDP holdtime: `R1(config)# lldp holdtime *seconds*`
- Configure the LLDP reinit timer: `R1(config)# lldp reinit *seconds*`



```
R1#show lldp traffic

LLDP traffic statistics:
  Total frames out: 4
  Total entries aged: 0
  Total frames in: 3
  Total frames received in error: 0
  Total frames discarded: 0
  Total TLVs discarded: 0
  Total TLVs unrecognized: 0
R1#
R1#show lldp interface

GigabitEthernet0/0:
  Tx: enabled
  Rx: enabled
  Tx state: IDLE
  Rx state: WAIT FOR FRAME

GigabitEthernet0/1:
  Tx: enabled
  Rx: enabled
  Tx state: IDLE
  Rx state: WAIT FOR FRAME

GigabitEthernet0/2:
  Tx: enabled
  Rx: enabled
  Tx state: INIT
  Rx state: WAIT PORT OPER

GigabitEthernet0/3:
  Tx: enabled
  Rx: enabled
  Tx state: INIT
  Rx state: WAIT PORT OPER
```

SHOW LLDP STATUS

```
R1#show lldp

Global LLDP Information:
  Status: ACTIVE
  LLDP advertisements are sent every 30 seconds
  LLDP hold time advertised is 120 seconds
  LLDP interface reinitialisation delay is 2 seconds
```

SHOW ALL LLDP NEIGHBORS

```
R1#show lldp neighbors
Capability codes:
  (R) Router, (B) Bridge, (T) Telephone, (C) DOCSIS Cable Device
  (W) WLAN Access Point, (P) Repeater, (S) Station, (O) Other

Device ID           Local Intf      Hold-time  Capability  Port ID
SW1                  Gi0/0           120        B           Gi0/0
R2                   Gi0/1           120        R           Gi0/0

Total entries displayed: 2
```

SHOW LLDP NEIGHBORS in DETAIL

```
R1#show lldp neighbors detail

Local Intf: Gi0/0
Chassis id: 0c04.41d2.1a00
Port id: Gi0/0
Port Description: GigabitEthernet0/0
System Name: SW1

System Description:
Cisco IOS Software, vios_l2 Software (vios_l2-ADVENTERPRISEK9-M), Version 15.2(4.0.55)E, TEST ENGINEERING ESTG_WEEKLY BUILD, synced to END_OF_FLO_ISP
Technical Support: http://www.cisco.com/techsupport
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Compl

Time remaining: 99 seconds
System Capabilities: B,R
Enabled Capabilities - not advertised
Management Addresses - not advertised
Auto Negotiation - not supported
Physical media capabilities - not advertised
Media Attachment Unit type - not advertised
Vlan ID: - not advertised
```

SHOW SPECIFIC LLDP DEVICE ENTRY

```
R1#show lldp entry SW1
```

```
Capability codes:  
  (R) Router, (B) Bridge, (T) Telephone, (C) DOCSIS Cable Device  
  (W) WLAN Access Point, (P) Repeater, (S) Station, (O) Other  
-----  
Local Intf: Gi0/0  
Chassis id: 0c04.41d2.1a00  
Port id: Gi0/0  
Port Description: GigabitEthernet0/0  
System Name: SW1  
  
System Description:  
Cisco IOS Software, vios_l2 Software (vios_l2-ADVENTERPRISEK9-M), Version 15.2(4.0.55)E, TEST ENGINEERING ESTG_WEEKLY BUILD, synced to END_OF_FLO_ISP  
Technical Support: http://www.cisco.com/techsupport  
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Compi  
  
Time remaining: 119 seconds  
System Capabilities: B,R  
Enabled Capabilities: R  
Management Addresses - not advertised  
Auto Negotiation - not supported  
Physical media capabilities - not advertised  
Media Attachment Unit type - not advertised  
Vlan ID: - not advertised
```



LLDP show commands summary

- **R1# show lldp**
→ shows basic information about LLDP (timers, version)
- **R1# show lldp traffic**
→ displays how many LLDP messages have been sent and received
- **R1# show lldp interface**
→ displays which interfaces LLDP tx/rx is enabled on
- **R1# show lldp neighbors**
→ lists LLDP neighbors and some basic information about each neighbor
- **R1# show lldp neighbors detail**
→ lists each LLDP neighbor with more detailed information
- **R1# show lldp entry name**
→ displays the same info as above, but for the specified neighbor only